Jay Niketan Pathare

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EDUCATION

Masters in Computer Science *August 2025 – December 2026 | Available January 2027*
University at Buffalo, The State University of New York, Buffalo, NY **GPA: 3.852 / 4.0**
Bachelor of Engineering, Information Technology
Vivekanand Education Society's Institute of Technology, Mumbai University, India *August 2019 – May 2023*

EXPERIENCE

Graduate Student Developer (CSE 611 Project) | Team of 5 | HeinOnline, Buffalo *January 2026 – April 2026*

- Engineered a fault-tolerant OCR-LLM pipeline using Tesseract-OCR and Qwen-35B across three redundant sources (CrossRef API, web scrapers, vision pipeline) to extract metadata from 1,000+ legacy law journals.
- Built a fuzzy-matching (rapidfuzz) combinator engine to merge asynchronous data streams and automate author metadata entry, reducing manual workload by 90% while sustaining 95%+ accuracy.
- Containerized the pipeline using Docker and validated outputs via automated evaluation against human-edited ground truth.

Software Engineer | Thesis Mumbai Tech (Startup), Mumbai, India *August 2024 – August 2025*

- Built and scaled healthcare modules (Patient Management, Consent) supporting 10k+ records, and designed an IoT baby-warmer system with real-time PostgreSQL pipelines and 2s live monitoring.
- Created a color-blindness diagnostic module achieving 97% clinical reliability, enabling automated report generation and real-time result visualization for medical practitioners.
- Containerized 3+ projects using Docker, reducing setup time by 90%, and integrated secure PDF generation and real-time video conferencing features.
- Conducted 20+ technical interviews, mentored new hires, and aligned with clients on UI/UX requirements.

Cloud Engineer Intern | Data Maven, Mumbai, India *November 2023 – May 2024*

- Designed and deployed a scalable cloud infrastructure on AWS using EC2 and RDS, optimizing VPC networking to ensure secure, low-latency communication for backend services.
- Streamlined resource provisioning, reducing setup time and supporting the deployment of data-intensive applications.

Data Engineer Intern | Go Digital Technology Consulting, Mumbai, India *June 2023 – August 2023*

- Cleaned and analyzed large-scale business datasets using Python (Pandas, NumPy) and MySQL, generating reports to highlight key operational trends.

TECHNICAL SKILLS

- **Languages:** Python, MySQL, PostgreSQL, HTML/CSS
- **Frameworks:** Pandas, NumPy, Django, Matplotlib
- **Artificial Intelligence & Machine Learning:** scikit-learn, PyTorch, TensorFlow, Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Graph Neural Networks (GNNs)
- **Software Engineering:** Object-Oriented Programming, REST API, Full-Stack Development, Backend Development, API Development, Unit Testing, ETL, LLM-based workflows, Prompt Engineering, AI-assisted automation
- **Tools:** Docker, Git, Tableau, OAuth/Auth0, WebRTC

ACADEMIC PROJECTS

Decoupled Learning Management System (LMS) | Individual Project *July 2026*

- Architected a full-stack LMS (Django REST Framework, React) with Redis-backed real-time messaging via Django Channels for live quizzes, leaderboards, and rate-limited chat; supported 45+ concurrent users with sub-500ms broadcast latency.
- Implemented idempotent server-side quiz grading, role-based access control for instructors/students, and a backtracking algorithm to generate conflict-free course schedules from section and term constraints.
- Secured the LMS with stateless JWT and custom WebSocket authentication; load-tested REST/WebSocket workloads with Locust and asyncio, reducing p95 REST latency 26% and quiz submission latency from 2000ms to 1500ms..

Fraud Detection using Graph Neural Networks | Team of 2 | University at Buffalo, NY *March 2026*

- Built a GraphSAGE-based fraud detection model on the IEEE-CIS dataset, utilizing Focal Loss to effectively handle a severe 27:1 class imbalance and achieving a 0.9259 AUC-ROC.
- Converted tabular transaction data into a complex graph network using shared card, device, and email features, capping node groups at 50 to prevent memory explosion while successfully capturing relational fraud rings.

Temperature Monitoring System | Team of 3 | University at Buffalo, NY *November 2025*

- Architected an end-to-end IoT data pipeline using Arduino microcontrollers to stream continuous telemetry data at 2-second intervals over WiFi to a ReactJS dashboard, achieving sub-second UI rendering latency.
- Implemented a Django REST backend and PostgreSQL database to process thousands of sensor logs with <50ms API response times, integrating Auth0 to secure 100% of data transmission endpoints.

CERTIFICATIONS, AWARDS & PUBLICATIONS

- **AWS Cloud Practitioner** *July 2024*
- **Certificate of Appreciation**: Event Manager, CSE Workshop "GitHub: Hands-On from Basics to Advanced," Dept. of Computer Science & Engineering, University at Buffalo. *February 2026*
- **Research Publication:** J. Pathare et al., "Music Genre Classification," IJRAR, Vol. 10, Issue 2, Apr. 2023. Achieved 97.68% accuracy via audio feature extraction using CatBoost and KNN. Link.